

Neural Induction and Factors That Stabilize a Neural Fate

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The neural ectoderm of vertebrates forms when the bone morphogenetic protein (BMP) signaling pathway is suppressed. Herein, we review the molecules that directly antagonize extracellular BMP and the signaling pathways that further contribute to reduce BMP activity in the neural ectoderm. Downstream of neural induction, a large number of “neural fate stabilizing” (NFS) transcription factors are expressed in the presumptive neural ectoderm, developing neural tube and ultimately in neural stem cells. Herein, we review what is known about their activities during normal development to maintain a neural fate and regulate neural differentiation. Further elucidation of how the NFS genes interact to regulate neural specification and differentiation should ultimately prove useful for regulating the expansion and differentiation of neural stem and progenitor cells. **Birth Defects Research (Part C) 87:249–262, 2009.** © 2009 Wiley-Liss, Inc.

Key words: BMP; FGF; Wnt; neural stem cells; transcriptional network; neural fate stabilization

INTRODUCTION

Nearly 100 years ago, Spemann and Mangold (1924) excised small bits of tissue from the point at which cells first involute in the amphibian gastrula and grafted these into the ventral belly region of host embryos. Surprisingly, a secondary body axis formed at the transplant site, and it included a neural tube and axial mesoderm (notochord, somites). By using explants and hosts from differently pigmented species so that the donor cells could be identified, they demonstrated that the transplanted piece, now known as Spemann’s “organizer,” did not give rise to neural ectoderm, but instead formed dorsal mesoderm. They reasoned that the trans-

planted cells signaled adjacent host cells to become the neural tissue of the secondary axis; thus, the phenomenon of neural induction was discovered. Equivalent organizing regions with neural inducing capacity are found in zebrafish (shield), chick (Henson’s node), and mouse (node) embryos (Beddington, 1994; Shih and Fraser, 1996; Smith and Schoenwolf, 1998; Boettger et al., 2001). Hence, neural induction is one of the primary steps in vertebrate neural development, and understanding how it is accomplished is of fundamental importance.

Since this landmark observation, numerous studies focused on identifying the inducing molecule(s) that are necessary to establish

neural fate in vertebrate embryos. In addition, a large number of transcription factors whose expression is initiated by neural inductive signaling were identified. How these factors cooperatively stabilize the neural fate of embryonic cells is only beginning to be appreciated. Although understanding the acquisition of neural fate during embryogenesis is exciting, it also appears that these same signaling factors and transcriptional regulators may be expressed in endogenous niches of neural stem cells in the adult brain. Therefore, the developmental program that specifies a neural fate in the embryo may prove to be critically important for coaxing neural stem cells to repopulate damaged tissue. This review will summarize the current state of knowledge of the process of neural induction, the genes involved in neural fate stabilization, and speculate on how this information may play a role in regulating the differentiation of neural stem cells.

NEURAL INDUCTION

The organizer forms in the dorsal mesoderm in response to secreted factors from a signaling center located in the adjacent presumptive dorsal endoderm, which in amphibians is called the Nieuwkoop Center (Nieuwkoop, 1967;

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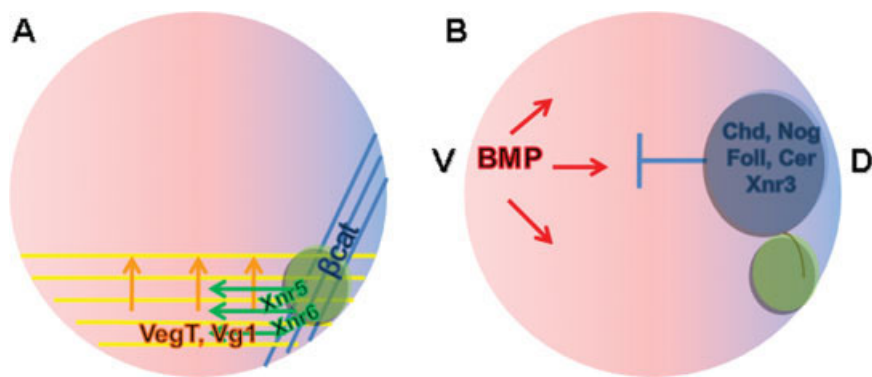


Figure 1. The dorsal-ventral axis is specified by the Nieuwkoop center and the Spemann organizer in *Xenopus*. (A) Before gastrulation, the dorsal axis is specified by the Nieuwkoop center (green circle), formed as a result of overlapping expression of VegT, Vg1, Nodals (Xnr), and β -catenin (β cat). (B) The Nieuwkoop center secretes dorsalizing molecules that induce the formation of the Spemann organizer (blue oval) in the dorsal mesoderm. The organizer expresses Chordin (Chd), Noggin (Nog), Follistatin (Foll), Cerberus (Cer), and in *Xenopus*, Nodal related-3 (Xnr3). These molecules actively antagonize BMP, which is secreted from the ventral side (red arrows). V, ventral; D, dorsal. Line intersecting the Nieuwkoop center represents the dorsal blastopore lip.

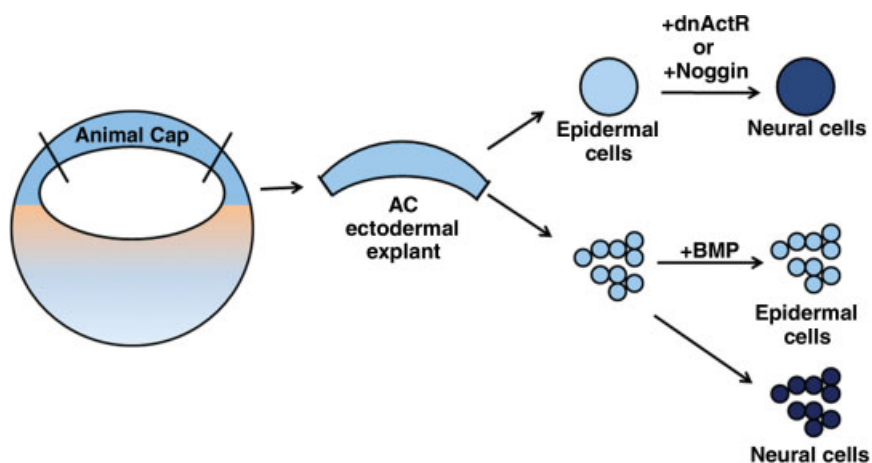


Figure 2. The default model of neural induction in *Xenopus*. An ectodermal explant dissected from the AC of a blastula stage embryo forms epidermis (light blue). If BMP signaling is inhibited in those cells by expressing a dominant-negative activin receptor (dnActR) or the extracellular antagonist Noggin, then neural tissue forms (dark blue). Dissociation of the explant into single cells without the addition of exogenous factors also leads to the formation of neural tissues. However, if exogenous BMP is added to the dispersed cells, they adopt an epidermal fate (light blue).

Gerhart et al., 1989; Agius et al., 2000; Vonica and Gumbiner, 2007). In *Xenopus*, the Nieuwkoop Center forms where the expression domains of VegT, Vg1, Nodal-related (Nr) proteins, and β -catenin overlap (see Fig. 1). The Spemann-Mangold experiment predicted that a neural-inducing substance emanated from the organizer. Although several studies sought to identify the neural

inducer(s), more recent work led to the hypothesis that ectoderm forms neural tissue not only by an activating signal but also by default (Sato and Sargent, 1989; Hawley et al., 1995; Wilson and Hemmati-Brivanlou, 1995; Weinstein and Hemmati-Brivanlou, 1997). Pieces of naïve embryonic ectoderm dissected from the animal pole of *Xenopus* blastulae (called animal cap [AC] ectoder-

mal explants) form non-neural epidermis when cultured without the addition of growth factors (see Fig. 2). However, if the AC ectodermal explant is first dissociated into single cells, the cells become neural, indicating that: (1) short range cell-to-cell signaling is required for the formation of epidermis, and (2) in the absence of this signaling, neural tissue forms by default (Sato and Sargent, 1989; Wilson and Hemmati-Brivanlou, 1995). In support of this hypothesis, experiments showed that the inhibition of transforming growth factor- β (TGF- β) signaling through the use of a dominant-negative activin receptor caused AC ectodermal explants to express neural marker genes (Hemmati-Brivanlou et al., 1992; Hemmati-Brivanlou and Melton, 1994; Hawley et al., 1995; Hemmati-Brivanlou and Thomsen, 1995). However, the dominant-negative activin receptor interferes with signaling of all of the members of the TGF- β family, including TGF- β , activin, Nr, and bone morphogenetic proteins (BMPs). Further work demonstrated that BMPs are the important ligands for neural versus epidermal fate induction.

BMP Signaling

Several studies pointed to the BMP signaling pathway as key for neural induction, but as a negative regulator and not as a positive inducer. First, using expression screening techniques to identify novel genes that induce a neural fate in AC ectodermal explants, secreted antagonists of BMPs were discovered. These proteins are secreted from the organizer (see Fig. 1), and include Noggin (Zimmerman et al., 1996), Chordin (Piccolo et al., 1996), Follistatin (Fainsod et al., 1997), Cerberus (Bouwmeester et al., 1996), and Xnr3 (Smith et al., 1995). They bind to BMP in the extracellular space and inhibit activation of its receptor (see Fig. 3); when this occurs in the naïve embryonic ectoderm, neural tissue forms instead of epidermis (Meinhardt, 2001; Vonica and Gumbiner,

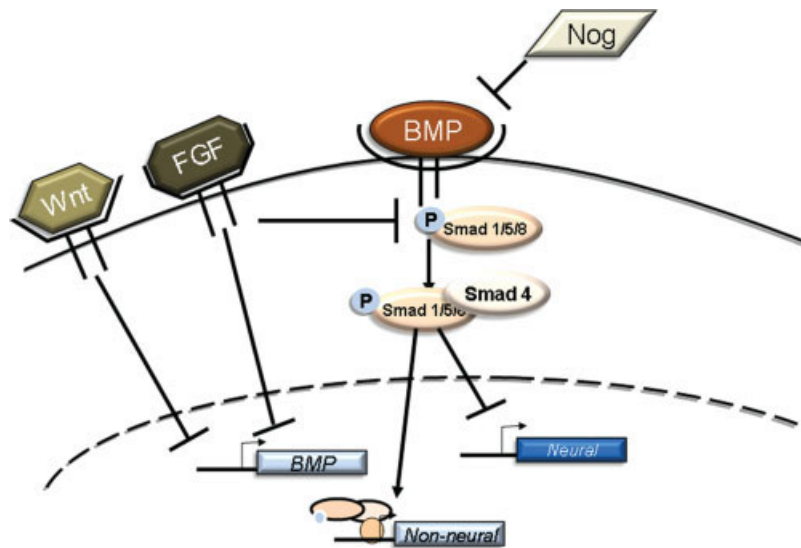


Figure 3. Pathways involved in neural induction. BMP2, 4, or 7 dimers bind to the serine-threonine receptor complex, leading to phosphorylation of the type I receptor by the type II receptor, which in turn phosphorylates (P) an appropriate receptor-regulated (R)-Smad (Smad1, 5, or 8). This phosphorylation enables the R-Smad to form a complex with the Common-Smad, Smad4, and this complex translocates to the nucleus to activate (non-neural) or repress (neural) target genes. The binding of secreted factors from the organizer, such as Noggin (Nog), to extracellular BMP prevents activation of its receptor complex. In addition, FGF and Wnt signaling lead to the inhibition of *bmp4* transcription. FGF signaling can also prevent the phosphorylation of Smads1/5/8.

2007). Second, when expression of one of these secreted proteins in the organizer is blocked, BMP signaling is maintained on the dorsal side of the embryo and the neural plate does not form (Wessely et al., 2004). Third, when dominant-negative BMP4 protein, dominant-negative BMP receptors, or any of the BMP antagonists are expressed in AC ectodermal explants, anterior neural genes are directly induced. Similarly, when these proteins are expressed on the ventral side of embryos, a secondary axis containing neural tissue forms (Hawley et al., 1995).

Mutational analysis and knock-down experiments in zebrafish, mouse, and frog embryos also demonstrate that BMP inhibition is required to form neural tissue. In zebrafish, double mutants for the BMP antagonists, *chordin* (a Chordin homolog) and *ogon* (a secreted Frizzled homolog), have a ventralized phenotype (Miller-Bertoglio et al., 1999; Yabe et al., 2003). *Xenopus tropicalis* embryos depleted of Chordin, Noggin, and

Follistatin protein have nearly a complete loss of the central nervous system (CNS) (Khokha et al., 2005). Mouse mutants lacking the BMP antagonists, Chordin (Bachiller et al., 2000), Noggin (McMahon et al., 1998), or Cerberus (Belo et al., 2000), express only posterior neural genes, and double mutants for Noggin and Chordin fail to develop anterior brain structures (Bachiller et al., 2000). These studies indicate that across vertebrates there is a conserved and necessary role for BMP inhibition in the induction of anterior neural tissue.

There are more than 30 known BMP proteins (Balemans and Van Hul, 2002), but it is BMP4 that is mainly involved in epidermal induction and neural inhibition (Sasai et al., 1995; Wilson and Hemmati-Brivanlou, 1995; Xu et al., 1995). Its expression is restricted to the non-neural ectoderm and ventral mesoderm, and overexpression ventralizes embryos (Fainsod et al., 1994; Hemmati-Brivanlou and Thomsen, 1995). A number of BMP target

genes have been identified and these include *Msx1* (Foerst-Potts and Sadler, 1997; Tucker et al., 1998; Feledy et al., 1999; Ishimura et al., 2000; Takeda et al., 2000; Yamamoto et al., 2000, 2001; Tribulo et al., 2003), *Msx2* (Foerst-Potts and Sadler, 1997), *Gata2* (Walmsley et al., 1994; Friedle and Knochel, 2002), *Vent1* (Gawantka et al., 1995; Friedle et al., 1998; Onichtchouk et al., 1998; Rastegar et al., 1999), and *Vent2* (Ladher et al., 1996; Onichtchouk et al., 1996, 1998; Rastegar et al., 1999; Trindade et al., 1999; Friedle and Knochel, 2002). *Msx2* is not likely to be involved in inhibiting the onset of expression of early neural genes because it is not expressed until the midgastrula stage, which is after neural induction occurs. In contrast, *Msx1*, *Gata2*, *Vent1*, and *Vent2* are expressed throughout the embryonic ectoderm of the animal pole of blastula embryos, and as gastrulation proceeds their expression domains are restricted to the ventral tissues in a pattern similar to that of *BMP4* (Hemmati-Brivanlou and Thomsen, 1995).

Gain-of-function studies indicate that *Msx1*, *Vent1*, and *Vent2* are involved in restricting the expression of neural genes to the dorsal ectoderm. Injection of *Msx1* mRNA inhibited neural tissue formation induced by Noggin (Ishimura et al., 2000), and overexpression of a dominant-activator form of *Msx1* prevented ventralization by BMP. However, although knock-down of *Msx1* function with antisense morpholino oligonucleotides (MOs) demonstrated that it is required for the ventralizing activity of BMP4, it is not required for epidermal development, inhibition of neural induction, or axis formation (Suzuki et al., 1997; Yamamoto et al., 2000). Knockdown experiments have not been reported for the Vent proteins, but overexpression of *Vent1* (Gawantka et al., 1995) or *Vent2* (Ladher et al., 1996; Onichtchouk et al., 1996; Friedle and Knochel, 2002) induces epidermis and inhibits the formation of both dorsal mesoderm and neural tissue.

Additionally, both Vent proteins are required to restrict expression of reporter constructs of *Geminin* (Taylor et al., 2006) and *Sox3* (Rogers et al., 2008), two early neural ectoderm transcription factors (discussed later), to the dorsal ectoderm.

Even though inhibition of BMP signaling is sufficient to induce neural tissue in AC ectodermal explants, it is not sufficient to induce it in the ventral, non-neural ectoderm of the embryo (Hawley et al., 1995; Wilson and Hemmati-Brivanlou, 1995; Rogers et al., 2008). For example, inhibition of BMP signaling in ventral ectoderm via overexpression of Smad6, an inhibitor of the BMP effector Smad1, repressed epidermis formation but failed to induce ectopic expression of neural markers (Delaune et al., 2005; Chang and Harland, 2007). It is possible that either a second downstream pathway needs to be silenced. For example, BMP inhibition is sufficient to induce neural tissue in ventral ectoderm if signaling through both Smad1 and Smad2 are inhibited (Chang and Harland, 2007). There is also evidence that additional instructive or permissive signals are required. Several experiments in frog and chick demonstrate that neural tissue is induced in non-neural ectoderm when BMP signaling is inhibited in the presence of fibroblast growth factor (FGF) (McMahon and Moon, 1989; Christian et al., 1991a; Rodriguez-Gallardo et al., 1997; Alvarez et al., 1998; Wilson et al., 2001; Delaune et al., 2005).

FGF Signaling

In frog, there are four known FGF receptors (FGFR1-4 α) implicated in neural development because of their expression in the ectoderm before neural induction and the subsequent restriction of their expression to neural tissue. Because *FGFR2* and *FGFR4 α* are upregulated at the onset of neural induction, they may play a specific role in the process (Hongo et al., 1999). To determine if FGF signaling is required for neural induc-

tion, signaling was inhibited by incubation in SU5402, a chemical inhibitor of FGF signaling (Mohammedi et al., 1997), or by overexpression of a dominant-negative FGFR1 receptor (XFD) (Amaya et al., 1991) or a dominant-negative FGFR4 α receptor ($\Delta 4\alpha$) (Hongo et al., 1999). XFD inhibits FGF3 and FGF4 signaling through FGFR1, FGFR2, and FGFR3, whereas $\Delta 4\alpha$ blocks FGF8 signaling through FGFR4 α (Ueno et al., 1992; Hongo et al., 1999; Bainter et al., 2001). XFD and SU5402 inhibited the induction of neural tissue in neuralized AC ectodermal explants as indicated by a loss of the pan-neural markers, *Ncam* and *Sox2* (Launay et al., 1996; Ribisi et al., 2000; Delaune et al., 2005). Overexpression of $\Delta 4\alpha$ led to a loss of anterior neural tissue (marked by *Nrp1*, *Bf1*, and *En2*) in whole embryos and in neuralized AC ectodermal explants (Hongo et al., 1999). Furthermore, although dissociated AC ectodermal explants are thought to be neuralized by the reduction in levels of BMP protein because of cell dispersal, recent studies showed that BMP signaling still occurs in these cultures (Hurtado and De Robertis, 2007), and that MAP kinase signaling may drive the conversion of cells to neurons (De Robertis and Kuroda, 2004; Kuroda et al., 2005). Although these studies suggest that FGF signaling is required for the formation of neural tissue in dissociated AC ectodermal explants, disruption of FGF signaling via the MAP kinase pathway by inhibiting Ras did not affect the induction of anterior neural genes, but inhibited the posteriorization of neural tissue in AC ectodermal explants (Ribisi et al., 2000). This is consistent with several previous studies indicating that FGF is required for the anterior-posterior patterning of the neural plate (reviewed in Gould and Grainger, 1997; Gamse and Sive, 2000). Moreover, treatment of embryos with SU5402 decreased some dorsal mesodermal markers, including *Noggin*; thus, the decrease in neural tissue may be accomplished indirectly by

reducing the amount of dorsal mesoderm available to induce it (Delaune et al., 2005). Taken together, these experiments demonstrate that signaling through FGFR4 α may be required for the induction of anterior neural tissue, and that FGFR1, FGFR2, and MAP kinase signaling are required for posteriorizing neural tissue.

Other studies demonstrate a role for FGF in neural induction; however, they do not demonstrate that the function of FGF is independent of its ability to inhibit *BMP* transcription or the BMP signaling pathway (Baker et al., 1999; Hongo et al., 1999; Ishimura et al., 2000; Wilson et al., 2000; Wilson et al., 2001; Pera et al., 2003; Sheng et al., 2003; Linker and Stern, 2004; Rentzsch et al., 2004; Wittler and Kessel, 2004; Delaune et al., 2005). In chick and zebrafish, FGF signaling inhibits *BMP* gene expression (Furthauer et al., 1997; Wilson et al., 2000), and in frog and zebrafish, FGF signaling inhibits BMP signaling through the phosphorylation and inactivation of its effector protein, Smad1 (Schier, 2001; Pera et al., 2003). In this manner, FGF may cooperate with or be redundant with the activity of extracellular BMP antagonists. Taken in total, these data support a role for FGF in patterning the anterior-posterior axis of neural tissues and in reinforcing the antagonism of the BMP pathway, but its requirement for neural induction independent of BMP inhibition remains in question (Lamb and Harland, 1995; Wilson et al., 2000; Takemoto et al., 2006; Rogers et al., 2008; Weisinger et al., 2008).

Wnt Signaling

Maternal Wnt signaling and the inhibition of zygotic Wnt are important for the formation of neural tissue. The establishment of the dorsal-ventral axis and the formation of the Nieuwkoop Center are dependent on the dorsal accumulation of the Wnt effector, β -catenin (Larabell et al., 1997). However, it is unclear if there is a specific role for Wnt signaling in

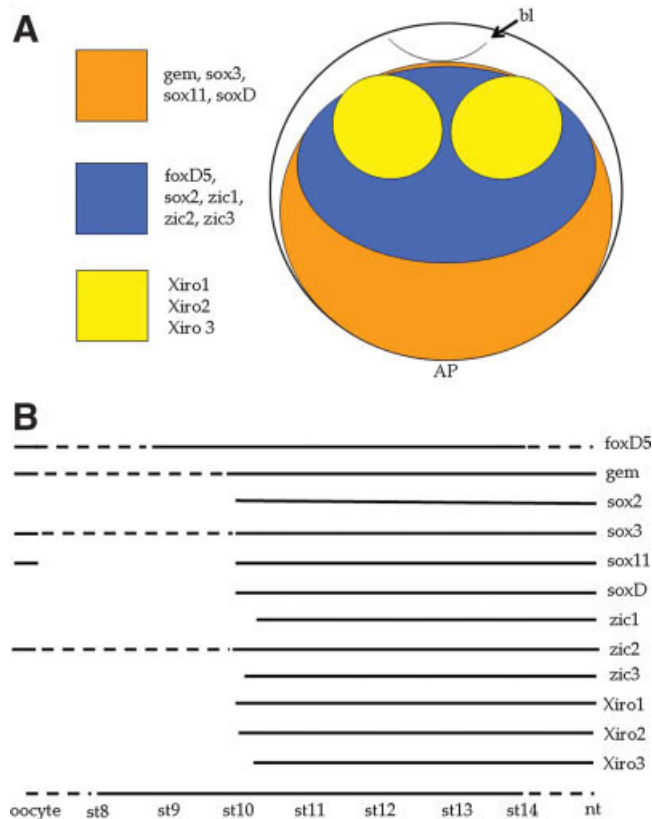


Figure 4. Twelve neural genes have overlapping expression patterns during the establishment of the *Xenopus* neural plate. **(A)** At the onset of gastrulation (st10), some NE genes are expressed throughout the dorsal ectoderm (orange: *geminin* [*gem*], *sox3*, *sox11*, *soxD*), some are expressed only in a broad dorsal band adjacent to the blastopore lip (bl) (blue: *foxD5*, *sox2*, *zic1-3*), and the *Xiro* genes (yellow) are expressed weakly in two posterior-lateral bands. Dorsal view with animal pole (AP) to the bottom. **(B)** The temporal expression patterns of these genes also overlap extensively. Four genes (*foxD5*, *gem*, *sox3*, and *zic2*) are expressed maternally (oocyte) with detectable mRNAs through blastula stages; *sox11* is expressed in oocytes but is not detected at blastula stages. The zygotic expression of most of these genes begins at late blastula (st9) to early gastrula (st10). The expression of nearly all is maintained through neural tube (nt) stages; in contrast, *foxD5* is lost as the neural plate is established (st14).

neural induction. Supporting a requirement for Wnt signaling in axis formation, overexpression of either Wnt1 or Wnt3a, which are expressed in neural tissue, induced a secondary axis (Steinbeisser et al., 1993). However, their ability to induce neural tissue in naïve ectoderm has not been tested. The strongest evidence for a role of Wnt signaling in neural induction was the experiments demonstrating the induction of *Ncam* in AC ectodermal explants in the absence of mesoderm formation in response to mRNA injections of either *Wnt8* or an activated form of β -catenin. Both inhibited *BMP4* expression, leading

to the proposal that Wnt signaling sensitizes AC ectodermal explants to extracellular BMP inhibition (Baker et al., 1999). However, it is worth noting that *Wnt8* normally is expressed in the ventral ectoderm, not the dorsal, presumptive neural ectoderm (Christian et al., 1991b), and in whole embryos it induces dorsal mesoderm (Otte and Moon, 1992), indicating an indirect effect. Also, loss of Wnt signaling by overexpression of a dominant-negative transcriptional effector (TCF3) or a dominant-negative receptor (Frizzled8) did not interfere with the induction of neural markers by Noggin (Rogers et al., 2008). Thus, there is little

evidence for a direct role of Wnt signaling in neural induction.

In fact, studies in both frog and chick support a requirement for the inhibition of Wnt signaling for neural induction. In frog, overexpression of a dominant-active form of β -catenin inhibited the expression of *Sox2* and *Sox3* in whole embryos and inhibited the induction of *Sox3* by Noggin in AC ectodermal explants (Heeg-Truesdell and Labonne, 2006). Similarly, overexpression of Wnt3a in chick embryos inhibited both the neuralization of ectoderm by BMP inhibition and FGF signaling (Wilson et al., 2001). However, like FGF, Wnt signaling does play an important role in the anterior-posterior patterning of the neural plate (Gould and Grainger, 1997; Gamse and Sive, 2000).

NEURAL FATE STABILIZATION

As the presumptive neural ectoderm is established via neural inductive signaling, a large number of transcription factors are coexpressed in broad overlapping domains (Fig. 4; Sasai, 1998; Moody and Je, 2002). The mRNAs of several of these genes (*Geminin*, *Sox3*, *Sox11*, *SoxD*) are found throughout the dorsal ectoderm at the onset of gastrulation. The transcripts of others (*FoxD5*, *Sox2*, *Zic1*, *Zic2*, *Zic3*) are concentrated in a broad band near the blastopore lip. The transcripts of a third group (*Xiro1*, *Xiro2*, *Xiro3*) are detected in two dorso-lateral bands near the blastopore lip. Although a few are expressed maternally, most are expressed around the onset of gastrulation and continue to be expressed in the neural tube (see Fig. 4). Experiments in *Xenopus* demonstrated that although most of these transcription factors do not induce ectopic neural tissue on their own, when they are overexpressed dorsally by mRNA injection, they all expand the neural plate. Previous studies propose that some of these genes modulate the competence of neural ectoderm to respond to signaling

molecules (*Sox2*, *Sox3*, *Zic1*, *Zic3*), some maintain an immature neural state (*Geminin*, *FoxD5*, *Zic2*), whereas others promote the expression of neural determination/differentiation bHLH genes (*SoxD*, *Sox11*, *Xiro1*, *Xiro2*, *Xiro3*). How these transcription factors relate to each other to stabilize a neural fate and regulate the progression from the initially induced neural ectoderm to neural progenitor cells has yet to be revealed.

Sox Genes

Sox genes are named for their similarity to the testis determination factor *Sry*, and a conserved high-mobility group (HMG) (Sry-related HMG-box) domain that confers their DNA binding ability (Gubbay et al., 1990; Sinclair et al., 1990; Wegner, 1999). They are transcription factors that bind to the minor groove of DNA, inducing a sharp bend and regulating gene transcription (Guth and Wegner, 2008). Of the 10 groups of *Sox* proteins, only *SoxD* (a *Xenopus* member of the *SoxG* group; Wegner, 1999), *SoxB* (*Sox1*, 2, 3, 14, 21), *SoxC* (*Sox4* and 11), and *SoxE* (*Sox8*, 9, 10) groups are involved in neural induction and development (Guth and Wegner, 2008).

Sox2 and *Sox3* are well documented early pan-neural markers (Uwanogho et al., 1995; Collignon et al., 1996; Penzel et al., 1997; Rex et al., 1997; Mizuseki et al., 1998a; Uchikawa et al., 1999; Graham et al., 2003; Linker and Stern, 2004; Takemoto et al., 2006; Rogers et al., 2008), and are part of the *SoxB1* (*Sox1*, *Sox2*, *Sox3*) subfamily of transcriptional activators (Penzel et al., 1997; Uchikawa et al., 1999). They are both expressed in neural progenitor cells throughout CNS development and are required for neural progenitor maintenance (Graham et al., 2003; Ellis et al., 2004; Pevny and Placzek, 2005; Rogers et al., 2009). Their expression patterns are similar but not identical (see Fig. 4). *Sox3* is expressed mater-

nally and is pan-ectodermal until the mid-gastrula stage when expression is restricted to the dorsal ectoderm (Penzel et al., 1997). In contrast, *Sox2* expression begins at the onset of neural induction only in the neural ectoderm (Nitta et al., 2006). Other differences include: *Sox3* is repressed in the floor plate of the neural tube before *Sox2*, expressed much more strongly in the otic placodes, and expressed in the developing lens, whereas *Sox2* is expressed in the retina (Penzel et al., 1997; Elkins and Henry, 2006; Nitta et al., 2006; Rogers et al., 2008). These differences in their spatiotemporal expression patterns indicate that they are regulated differently, and promoter studies and bioinformatic analysis indicate that this is true (Uchikawa et al., 2003; Rogers et al., 2008). The *Sox2* regulatory region has a neural induction module that is conserved across species, and in chick this module responds to both FGF and Wnt signaling for activation in the posterior neural plate (Takemoto et al., 2006). This module is present upstream of *Xenopus tropicalis Sox2*, but is not in the flanking sequence of *Sox3*. Thus far, reporter studies in transgenic frog embryos reveal that *Xenopus Sox3* expression is induced by the inhibition of BMP signaling and is restricted to the neural plate by the BMP targets, *Vent1* and *Vent2* (Rogers et al., 2008).

Rescue experiments indicate that the *SoxB1* proteins are redundant in tissues in which they are coexpressed. In mouse, loss of *Sox2* in the CNS is compensated by the upregulation of *Sox3* expression. Neurogenesis is decreased in these mice, but neural stem cells retain multipotency (Miyagi et al., 2008). In chick, overexpression of *Sox1* rescues the loss of *Sox2* (Graham et al., 2003). Although the *SoxB1* proteins can compensate for each other, each gene is also expressed in unique tissues during development and loss in that tissue can result in severe phenotypes. Mice mutants for either *Sox2* or *Sox3*

have relatively normal CNS formation (Malas et al., 2003; Taranova et al., 2006). However, loss of *Sox2* results in diminished neurogenesis in the mouse retina (Ferri et al., 2004) and anophthalmia in humans (Fantès et al., 2003). *Sox3*-null mice have craniofacial abnormalities and defective pituitary development (Rizzoti et al., 2004; Rizzoti and Lovell-Badge, 2007), and *Sox1*-null mice suffer from epilepsy because of complete loss of neurons in the ventral striatum (Malas et al., 2003; Ekonomou et al., 2005).

In early neural development, vertebrate *SoxB1* proteins function as transcriptional activators (Uchikawa et al., 1999; Bylund et al., 2003; Graham et al., 2003;). It has been proposed that they counteract neuronal differentiation induced by the *SoxB2* subfamily of repressor proteins (*Sox14* and *Sox21*) (Bylund et al., 2003). In this way, the two subgroups of *SoxB* proteins together may maintain a balance between proliferation and differentiation. In concordance, overexpression of both *Sox2* and *Sox3* in frog and zebrafish leads to a loss of neurogenesis in the cranial placodes (Dee et al., 2008; Schlosser et al., 2008) and expansion of neural progenitors in the frog and chick neural tube at the expense of epidermal development and neuronal differentiation (Kishi et al., 2000; Graham et al., 2003; Rogers et al., 2009).

The *SoxC* gene, *Sox11*, is expressed in the developing nervous system of chick (Uwanogho et al., 1995), mouse (Hargrave et al., 1997), zebrafish (de Martino et al., 2000), and frog (Hyodo-Miura et al., 2002; Bruggmann et al., 2004). *Sox11* is induced in response to BMP inhibition by *Chordin* (Hyodo-Miura et al., 2002) and by ectopic expression of the *FoxD5* transcription factor (Yan et al., 2009). In *Xenopus* AC assays, *Sox11* expression induces pan-neural (*Ncam*) and anterior neural (*Otx2*, *En2*) markers through an interaction with the MAP kinase, NLK (Hyodo-Miura et al., 2002). *Sox11* is thought to be functionally

redundant with the other SoxC proteins, Sox4 and Sox12, because of their overlapping expression patterns, ability to bind to the same DNA gene sequence, and weak single gene knockout phenotypes in mouse (Dy et al., 2008; Hoser et al., 2008). In chick, although Sox11 is expressed in the neural epithelium, it is upregulated in maturing neurons (Uwanogho et al., 1995) and functions in the differentiation of neural progenitors into neurons (Bergsland et al., 2006).

SoxD, a member of the SoxG group, is unique to amphibians (Mizuseki et al., 1998b; Guth and Wegner, 2008). It is first detected at late blastula stages throughout the embryonic ectoderm, and like Sox3, it is then restricted to the dorsal ectoderm during gastrulation. *SoxD* is induced in response to the inhibition of BMP signaling, and like Sox2, can be induced in AC ectodermal explants by *Zic1*. Little is known about the requirement for FGF or Wnt signaling for the expression of *SoxD*, although the MEK5/ERK5 pathway has been implicated in its neuralizing activity (Nishimoto et al., 2005). Overexpression of *SoxD* in whole embryos expanded neural progenitors marked by *Nrp1*, induced the neural determination bHLH gene *Ngnr1*, and inhibited epidermal gene expression. Inhibition of *SoxD* by a dominant-negative construct suppressed neural tissues, a phenotype that was not rescued by Sox2, indicating that *SoxD* acts downstream (Mizuseki et al., 1998b).

Geminin

Geminin is a novel coiled-coil protein whose mRNA is expressed in a similar pattern as Sox3 (see Fig. 4); it is maternally expressed and ubiquitous in the animal pole ectoderm until gastrulation when it is first enriched dorsally, and then restricted to the neural ectoderm (Kroll et al., 1998). After gastrulation, *Geminin* expression is pan-neural with an anterior bias and encompasses a region wider than that of either Sox2 or Sox3

(Kroll et al., 1998; Kroll, 2007). *Geminin* expression can be induced by inhibiting BMP signaling, and it is maintained in the proliferative regions of the developing nervous system (Kroll, 2007). Geminin maintains a neural progenitor population by inhibiting reinitiation of DNA replication, thereby maintaining chromosomal integrity and preventing cell cycle exit (Seo and Kroll, 2006). The latter is controlled by an antagonistic interaction with the catalytic subunits of the SWI/SNF complex, Brg1 and Brahma (McGarry and Kirschner, 1998; Seo et al., 2005). Overexpression of *Geminin* inhibits epidermal development and neuronal differentiation, and expands the neural progenitor population (Kroll et al., 1998; Papanayotou et al., 2008). It has been suggested that *Geminin* maintains cell cycling during neural plate stages and represses various aspects of differentiation, allowing for the expansion of neural ectoderm (Luo and Kessel, 2004; Pitulescu et al., 2005; Seo and Kroll, 2006).

Experiments using a human *Geminin* regulatory region driving GFP expression indicated that as for Sox3, Vent1 and Vent2 restrict *Geminin* to the dorsal side of the embryo (Kroll et al., 1998). Dorsal expression of this *Geminin* reporter construct required positive regulation by the Wnt signaling effector, TCF3 (Taylor et al., 2006). Additionally, recent experiments demonstrated that *Geminin* expression is activated directly by two other early neural proteins, Sox3 (Rogers et al., 2009) and FoxD5 (Yan et al., 2009).

FoxD5

FoxD5 is a member of the forkhead/winged helix family of transcription factors (Solter et al., 1999; Fetka et al., 2000; Sullivan et al., 2001). Although there are homologous genes in zebrafish (*fkf8*; Odenthal and Nusslein-Volhard, 1998), mouse (*FoxD4/Fkh2*; Kaestner et al., 1995), and human (*FOXD5* [2q13]; Katoh and Katoh, 2004), no functional information is

currently available for any of these. In *Xenopus*, *FoxD5* maternal transcripts are localized to the animal pole, similar to those of Sox3 and *Geminin*. However, at gastrulation, *FoxD5* zygotic transcripts are more restricted than those of Sox3 and *Geminin* in the presumptive neural ectoderm (see Fig. 4). Unlike the other early neural transcription factor genes, *FoxD5* expression is not maintained throughout neural development, but is extinguished as the neural folds elevate and fuse, except at the midbrain/hindbrain junction and the tail bud. Also, unlike the other neural genes, *FoxD5* is only weakly induced by Noggin, not induced by Chordin, but is induced strongly by the maternally regulated Siamois pathway via Cerberus (Sullivan et al., 2001).

It is proposed that FoxD5, like Geminin and the SoxB1 proteins, functions to maintain a proliferating neural progenitor population. Dorsal injections of *FoxD5* mRNA expanded expression of other early neural genes (*Sox2*, *Sox3*, *Otx2*), and repressed neural patterning (*En2*, *Krox20*) and neural differentiation (*Ngnr1*, *NeuroD*, *Ntub*) genes (Sullivan et al., 2001). Although expression of *FoxD5* induced several neural genes in AC ectodermal explants (Sullivan et al., 2001), it only induces a few of these in the ventral ectoderm of embryos (Yan et al., 2009), suggesting that it acts downstream of neural induction. Microinjections of *FoxD5* dominant-activator and dominant-repressor constructs revealed that FoxD5 functions as a transcriptional repressor to expand the neural plate and deletion analysis revealed that it is dependent on the C-terminal domain of the protein. These results indicate that *FoxD5* contributes to maintaining an undifferentiated neural ectoderm during the early steps of neural plate formation.

Zic Genes

Three *Zic* genes (*Zic1*, *Zic2*, and *Zic3*), members of the zinc finger family of transcription factors, are

expressed in the early neural ectoderm (see Fig. 4). All three are expressed in the presumptive neural ectoderm during early gastrulation, and later become restricted to the lateral margins of the neural plate in mouse, fish, and frog (reviewed in Moody and Je, 2002). In *Xenopus*, *Zic1* and *Zic3* are transcribed 30–60 min after the *Chordin* gene, and can be induced in AC ectodermal explants by BMP antagonism (Nakata et al., 1997; Kuo et al., 1998; Mizuseki et al., 1998a; Nakata et al., 1998), indicating that they are likely transcribed in immediate response to neural induction. Expression of *Zic1* allows ectoderm to be more sensitive to neural induction by Noggin (Kuo et al., 1998), confirming its role in neural fate stabilization. Overexpression of *Zic1* and *Zic3* expands the neural plate and neural crest, and concomitantly represses epidermal fate. In AC ectodermal explants, *Zic1* activates other early neural markers (*Nrp1*, *SoxD*) in the absence of mesoderm induction. *Zic1* and *Zic3* both expand the expression of neural determination/differentiation bHLH genes (Mizuseki et al., 1998a), suggesting that they promote the transition to neural differentiation.

Zic2 is expressed in a similar pattern to *Zic1* and *Zic3* (Brewster et al., 1998; Nakata et al., 1998), and it similarly expands the neural plate and neural crest, and reduces epidermal markers (Brewster et al., 1998). In contrast, *Zic2* overexpression represses neural determination/differentiation bHLH genes, and can counteract the formation of ectopic neurons produced by *Ngfr1* mRNA injection. These studies suggest that during neural plate formation, *Zic1* and *Zic3* promote the onset of neural differentiation, whereas *Zic2* maintains cells in a more immature progenitor state.

Iroquois Genes

Genes within the *Drosophila* *Iroquois* complex (*Iro/Irx*) encode homeodomain proteins that are required for the activation of pro-

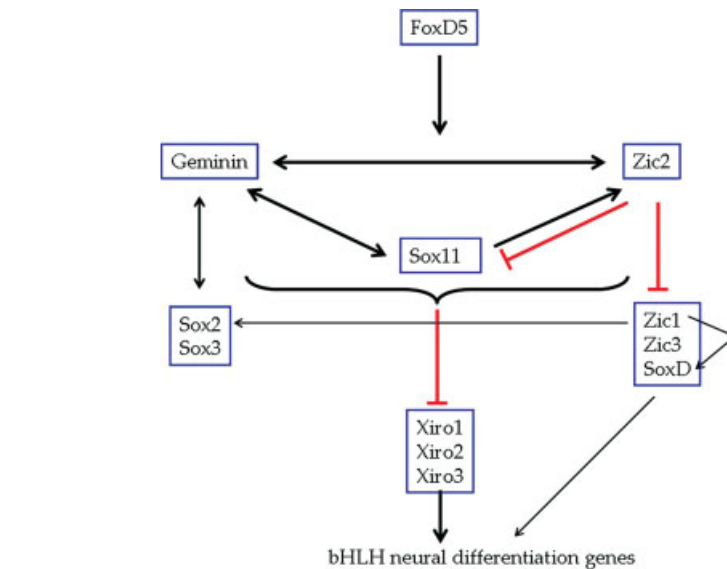


Figure 5. A summary of the data discussed in the text predicts a gene regulatory network that results in the stabilization of neural fate. After these 12 NFS genes are induced, FoxD5 upregulates *Geminin*, *Sox11*, and *Zic2* expression. These three genes regulate each other and together (bracket) they differentially affect the expression of the other eight NFS genes. *Geminin* and *Sox2/Sox3* directly regulate each other, and increased *Zic1* upregulates *Sox2* and *SoxD*. FoxD5, *Geminin*, *Sox11*, and *Zic2* maintain neural ectodermal cells in an immature, stem-like state, whereas *Zic1*, *Zic3*, *SoxD*, and *Xiro1-3* promote the onset of neural differentiation.

neural bHLH genes (Gomez-Skarmeta et al., 1996; Gomez-Skarmeta and Modollet, 2002). Several vertebrate homologues have been cloned, three of which in *Xenopus* (*Xiro1*, *Xiro2*, and *Xiro3*) are expressed in restricted bands in the neural ectoderm (see Fig. 4) just before the earliest expressed neural determination bHLH genes (*Ngfr1* and *NeuroD*). Later in development, *Xiro1-3* are expressed only in the dorsolateral neural tube and neural crest. All three *Xiro* genes are induced only weakly by Noggin, but their expression is greatly enhanced by the addition of posteriorizing factors, such as Wnt and FGF. These data indicate that they are expressed downstream of the aforementioned early neural genes, which do not require posteriorizing factors for their induction. In *Xenopus*, overexpression of each of the three *Xiro* genes causes the neural plate to expand and promotes the onset of neural differentiation (Bellefroid et al., 1998; Gomez-Skarmeta et al., 1998), but suppresses terminal differentiation into neurons (de la

Calle-Mustienes et al., 2002). Thus, these genes are predicted to function in a position intermediate between the aforementioned early neural genes and the neural determination/differentiation bHLH genes (see Fig. 5).

Do Early Neural Transcription Factors Interact to Regulate Neural Fate Stabilization?

Because of their overlapping expression patterns and their common overexpression phenotype of overexpanding the neural plate, it seems likely that together these early neural genes coordinately stabilize the newly induced neural fate, regulate neural plate formation, and control the onset of neural differentiation. However, our understanding of how this is accomplished is woefully incomplete. Several studies indicate that one aspect of stabilizing neural fate is by modifying the signaling pathways involved in neural induction. For example: (1) in the presence of *Zic1*, a lower concentration of Noggin is required to induce neural ectoderm (Kuo

et al., 1998); (2) *Sox11* induces neural marker genes by antagonizing Wnt signaling (Hyodo-Miura et al., 2002); and (3) *Geminin*, *Sox3*, and *Xiro1* each antagonize some aspect of the BMP4 signaling pathway (Kroll et al., 1998; Glavic et al., 2001; Gomez-Skarmeta et al., 2001; Rogers et al., 2008), and *FoxD5* reduces the nuclear localization of the BMP signaling effectors, phosphorylated SMAD1/5/8, reduces the ventral epidermal expression of BMP4 target genes, and upregulates the expression of *Szl*, a secreted BMP antagonist (Yan et al., 2009). Thus, one aspect of neural fate stabilization appears to be the regulation of the strength of signaling pathways, in particular that of BMP4, in the neural ectoderm. However, we know virtually nothing about how these several early neural genes, henceforth called neural fate stabilizing (NFS) genes, may cooperatively cause these effects.

The overexpression studies reviewed earlier indicate that the NFS genes can be placed into two functional groups: those that promote an immature, undifferentiated neural state (*Geminin*, *Sox2*, *Sox3*, *FoxD5*, *Zic2*), and those that promote the onset of neural differentiation (*SoxD*, *Sox11*, *Zic1*, *Zic3*, *Xiro1*, *Xiro2*, *Xiro3*). Are the genes within each group simply redundant, or do they have distinct roles in the transition from neural induction to neural differentiation? Answers to these questions are only just beginning to be experimentally addressed.

Recently, as a first step in putting the NFS genes into transcriptional order, we found that when endogenous *FoxD5* levels in the neural ectoderm were reduced by targeted injection of anti-sense MOs, the expression of each of the 11 other NFS genes was downregulated (Yan et al., 2009). This suggests that *FoxD5* is necessary for either the induction or the maintenance of their expression. As expected from previous studies (Sullivan et al., 2001), overexpression of *FoxD5* expanded the neural plate and thus the domains of each of the 11 other NFS genes;

this was a cell nonautonomous effect because the neural plate was expanded equally well in regions devoid of the *FoxD5*-expressing cells, which were marked by a lineage tracer. However, when only the lineage-tagged cells expressing elevated levels of *FoxD5* were examined, several responses were observed: (1) some genes that promote an immature neural fate (*Geminin*, *Zic2*) were upregulated; (2) *Sox11*, which previously had been characterized as promoting neural differentiation (Bergsland et al., 2006), also was upregulated; (3) some genes that promote an immature neural fate but are required for neural differentiation (*Sox2*, *Sox3*) were transiently and weakly repressed; and (4) genes that promote the onset of neural differentiation (*Zic1*, *Zic3*, *SoxD*, *Xiro1*, *Xiro2*, *Xiro3*) were strongly repressed. Thus, *FoxD5* appears to maintain an immature neural state by differentially regulating genes that either repress or promote neural differentiation. Further experiments demonstrated that *FoxD5* directly regulates *Geminin*, *Sox11*, and *Zic2* transcription, and that these three genes together phenocopy *FoxD5* overexpression and rescue *FoxD5* MO-mediated knockdown phenotypes.

These results indicate that the NFS genes coordinately stabilize neural fate and regulate the initiation of neural differentiation, and data from several laboratories predict a NFS gene regulatory network (see Fig. 5). *FoxD5* acts upstream of the other NFS genes, primarily by regulating *Geminin*, *Sox11*, and *Zic2*. This transcriptional triad cooperatively carries out most of the effects of *FoxD5* on the other NFS genes. They also regulate each other's expression, indicating a network rather than a linear path of transcriptional regulation. There are also mutual interactions between *Geminin* and *SoxB1* genes. Chick *Geminin* interacts in a complex at the N2 enhancer to promote *Sox2* transcription (Papanayotou et al., 2008), and frog *Sox2* and *Sox3* directly regulate *Geminin* expression (Rogers et al.,

2009). As mentioned earlier, *Zic1* upregulates both *Sox2* and *SoxD* expression. In addition, those genes that promote the onset of neural differentiation (*Zic1*, *Zic3*, *SoxD*, *Xiro1*, *Xiro2*, *Xiro3*) feedback to repress *FoxD5* expression, perhaps to allow the neural ectodermal cells to initiate differentiation (Yan et al., 2009). These studies are just the beginning of defining the regulatory network that promotes the transition from neural induction to differentiation. Much more work is needed to fully understand the gene interactions that comprise this network, and to determine what other factors are involved.

HOW MIGHT NFS GENES REGULATE NEURAL STEM CELLS?

Stem cell therapy is expected to have widespread benefits to patients suffering from neurodegenerative diseases and neuronal loss due to trauma, stroke, and congenital disease. Although the mature CNS harbors niches of self-renewing neural stem cells (Doetsch et al., 1999; Alvarez-Buylla et al., 2001), these make a significant contribution to only a few areas of brain (Cameron and McKay, 2001; Pencea et al., 2001). Therefore, many laboratories world-wide are researching how to manipulate different sources of stem cells to produce specific types of neurons for therapeutic treatments. It is most exciting that many of the NFS genes have already been implicated in controlling the differentiation of neural stem and progenitor cells. *Sox2* and *Sox3* are necessary for neural differentiation (Kishi et al., 2000; Wegner and Stolt, 2005; Wang et al., 2006). Each maintains neural stem/progenitor cells in a proliferative state upstream of neuronal terminal differentiation genes (Li et al., 1998; Zappone et al., 2000; Bylund et al., 2003; Graham et al., 2003; Ellis et al., 2004; Chung et al., 2006; Wang et al., 2006). *Sox2* is required for adult neural stem cell maintenance, and its loss causes cells to

leave the ventricular zone and exit the cell cycle (Graham et al., 2003; Ellis et al., 2004; Episkopou, 2005). In contrast, *Sox1* and *Sox3* appear to regulate neural progenitor cell states (Kan et al., 2004; Barraud et al., 2005; Chung et al., 2006; Wang et al., 2006). *Sox11* is reported to be upregulated as neural stem cells transition to neural progenitor cells, and later it maintains pan-neural genes in neuronal progenitors downstream of bHLH differentiation factors (Uwanogho et al., 1995; Wegner and Stolt, 2005; Bergsland et al., 2006). Thus, together these *Sox* genes may function downstream of *FoxD5*, *Geminin*, and *Zic2* to promote the initial steps from neural stem to neural progenitor cell. *FoxD5* transiently represses *Sox2*, *Sox3*, and *Sox11* (Yan et al., 2009), suggesting that it may act to delay the stem-to-progenitor transition. *Zic* genes, which promote the expression of neural determination/differentiation bHLH factors, are expressed in proliferative neural progenitor cells and are required for their expansion (Aruga et al., 2002; Ebert et al., 2003; Inoue et al., 2007). *Iro/Irx* genes are also involved in the initiation of expression of the neural determination/differentiation bHLH factors. Later, they play important roles in patterning the CNS (Rodriguez-Sequel et al., 2009; Stedman et al., 2009) and appear to regulate the number of cranial placode progenitors (Feijoo et al., 2009).

CONCLUSIONS

Much has been learned about the endogenous signaling pathways and transcriptional factors that induce the neural ectoderm and stabilize its neural fate. However, new experimental tactics will be needed to sort out how the various signaling pathways intersect to produce a neural fate, and likewise how the several NFS transcription factors regulate one another. Ultimately, producing neurally committed stem and progenitor cells in sufficient quantities and at the

appropriate state of differentiation for clinical applications requires that we understand the molecular mechanisms that regulate the normal process of *in vivo* neural development in greater detail.

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